Bazel

Bazel for C++ and Python

[Incremental Build]

HITESH KUMAR SHARMA
Technical Instructor & Consultant

NobleProg

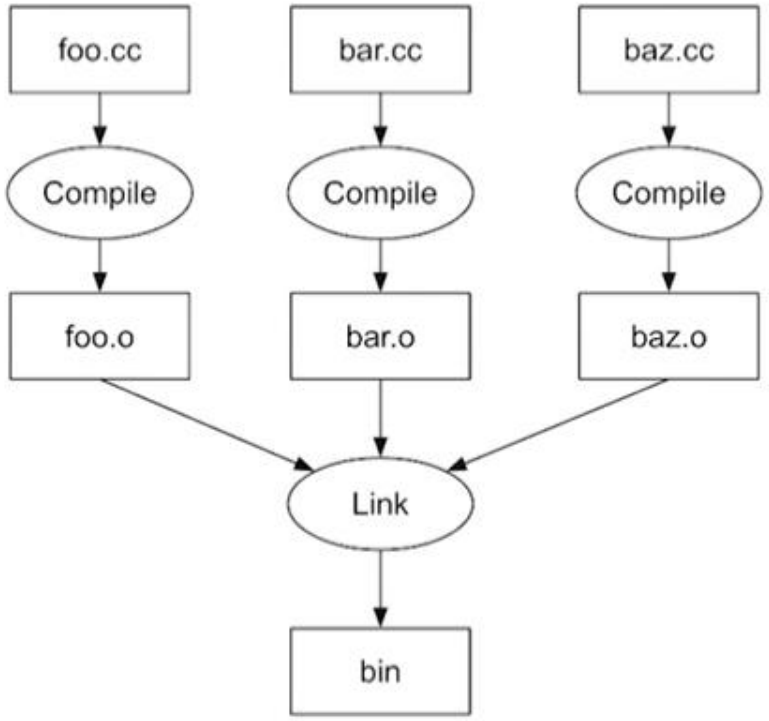
Bazel

Action

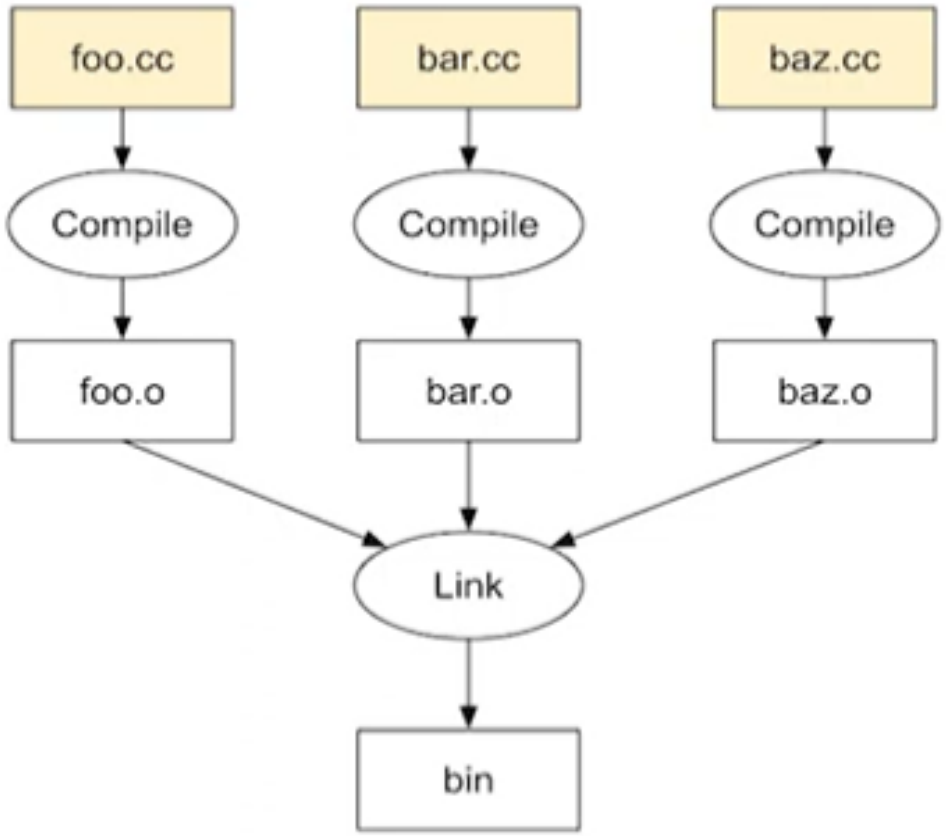
- A build is a collection of actions which can be represented by a graph
- An action runs a program on inputs and produces outputs
- Inputs are either source files or outputs produced by other actions

Bazel

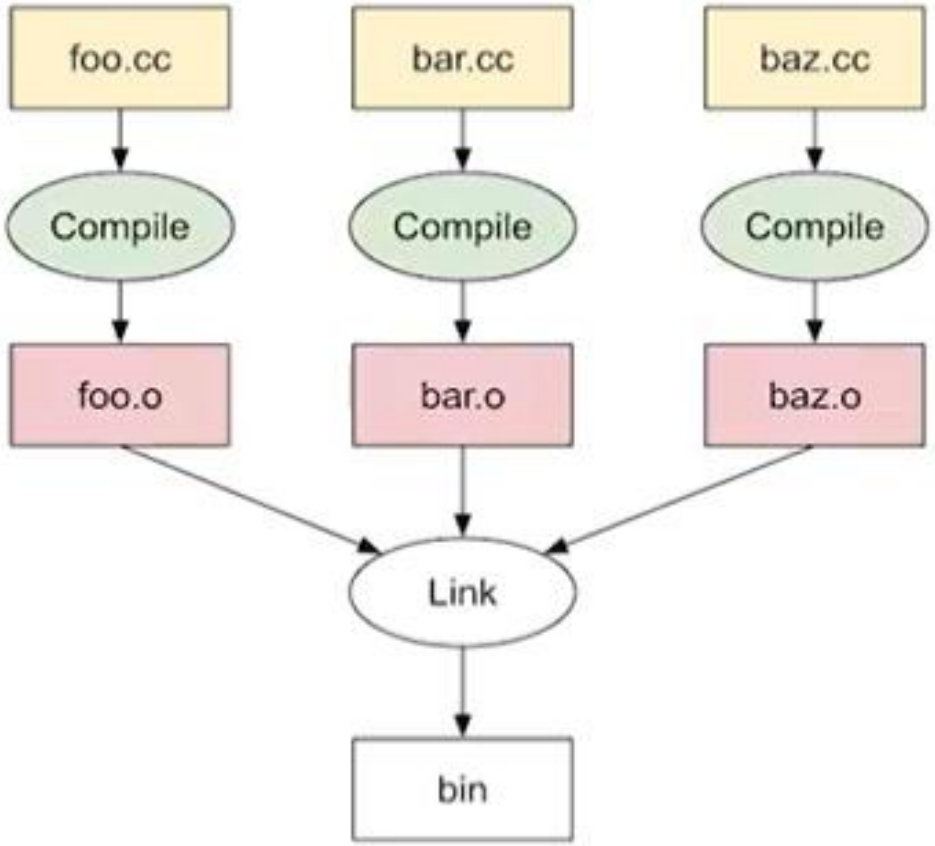
Action Graph



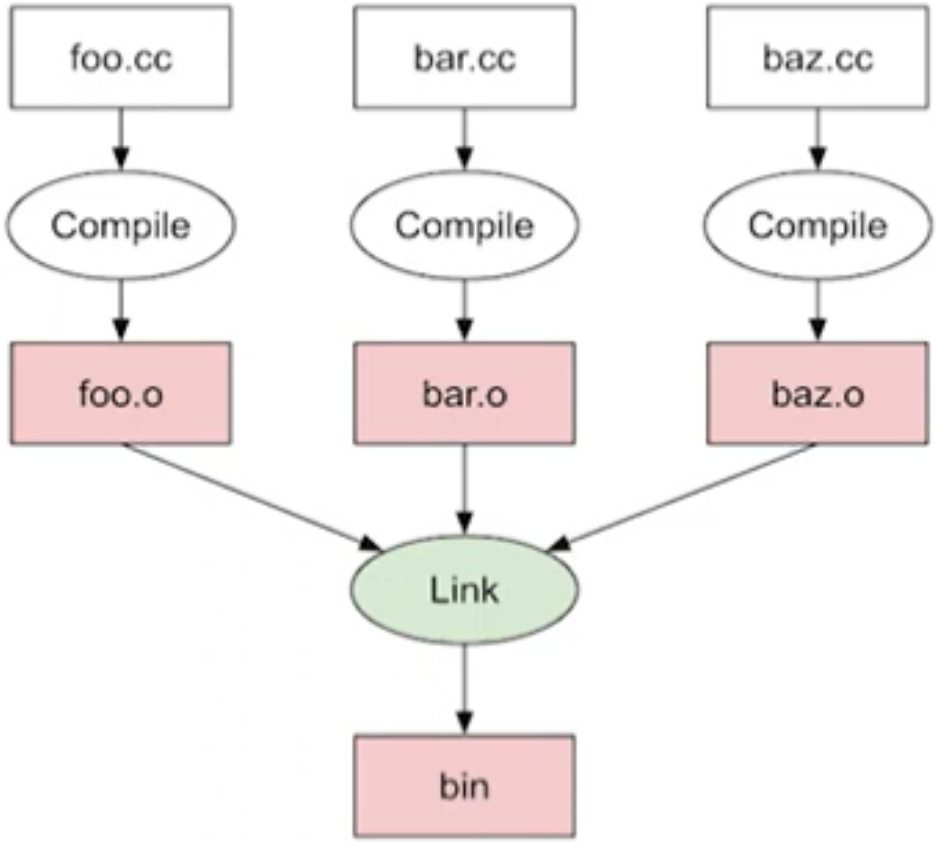
Action Graph



Action Graph



Action Graph



Bazel

Workspace

```
chiwang@chiwang ~/d/bazel-day-demo> tree .
.
├── bar.cc
├── baz.cc
├── BUILD
├── foo.cc
└── WORKSPACE

0 directories, 5 files
chiwang@chiwang ~/d/bazel-day-demo>
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Bazel

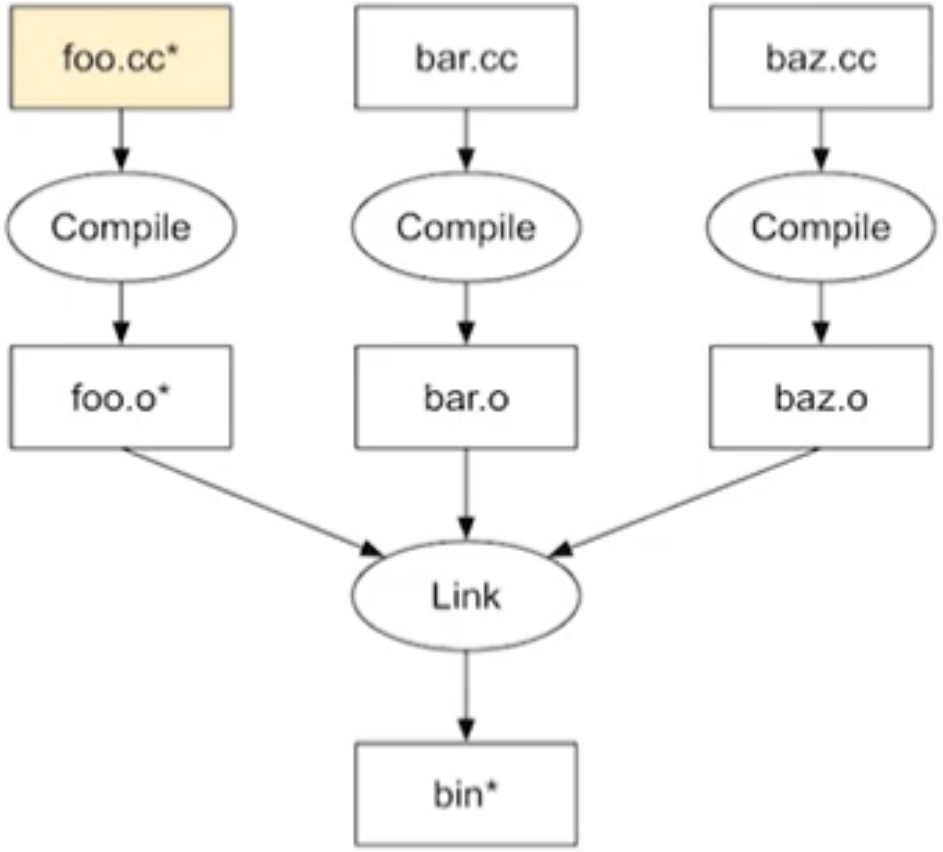
Demo: Clean build

```
Welcome to fish, the friendly interactive shell
Type help for instructions on how to use fish
chiwang@chiwang ~/d/bazel-day-demo> bazel build :bin
Starting local Bazel server and connecting to it...
INFO: Analyzed target //:bin (4 packages loaded, 12 targets configured).
INFO: Found 1 target...
Target //:bin up-to-date:
  bazel-bin/bin
INFO: Elapsed time: 6.583s, Critical Path: 4.62s
INFO: 5 processes: 1 internal, 4 linux-sandbox.
INFO: Build completed successfully, 5 total actions
chiwang@chiwang ~/d/bazel-day-demo>
```

Incremental build

- Bazel achieves speed by reusing previous build results.
- When you modify a source file, Bazel only runs the affected actions.

Incremental build



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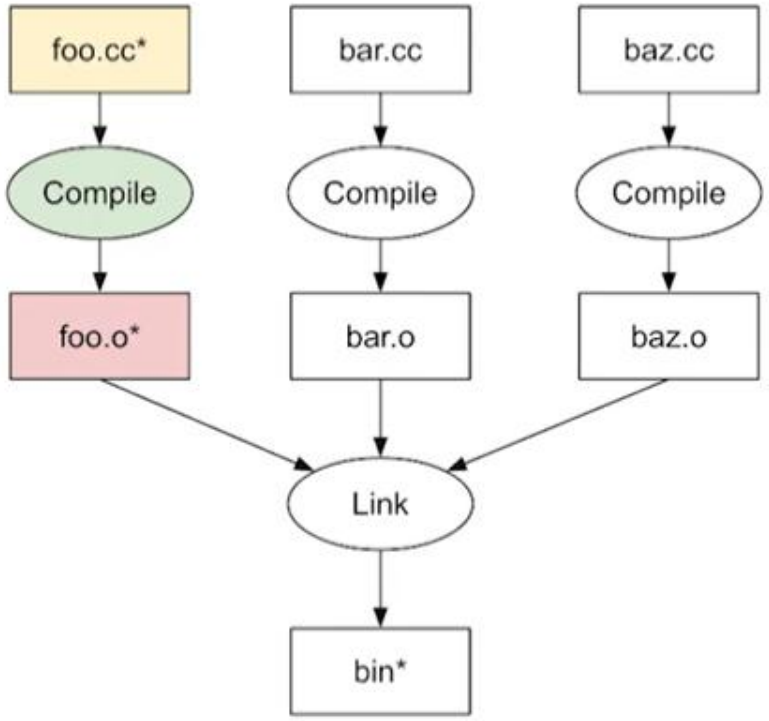
Incremental build



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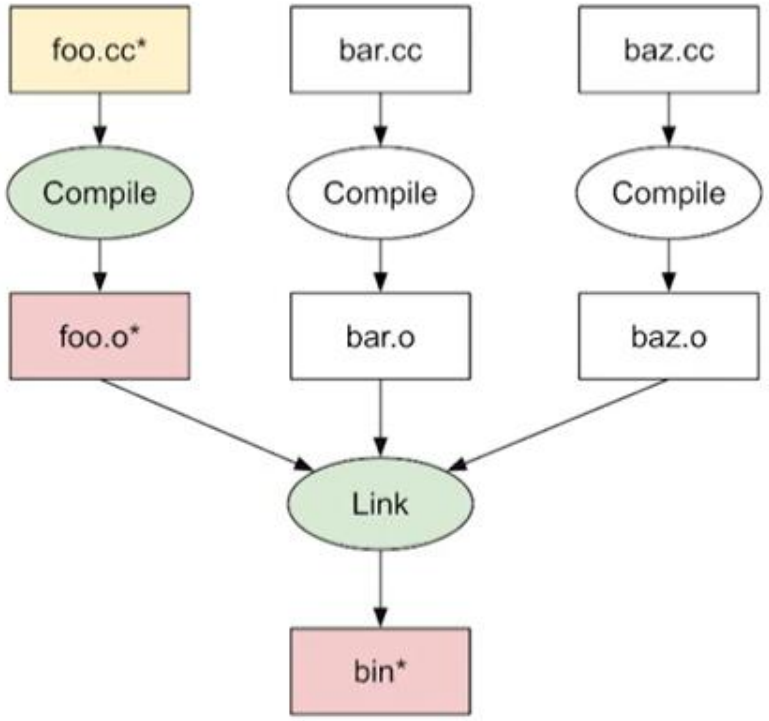
Incremental build



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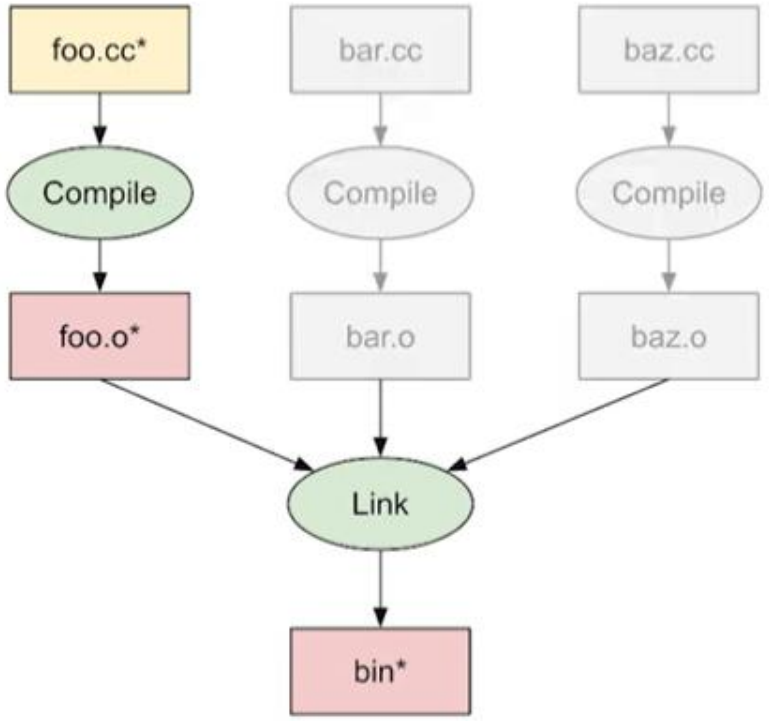
Incremental build



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Demo: Incremental build

```
Welcome to fish, the friendly interactive shell
Type help for instructions on how to use fish
chiwang@chiwang ~/d/bazel-day-demo> nvim foo.cc
chiwang@chiwang ~/d/bazel-day-demo> bazel build :bin
INFO: Analyzed target //:bin (0 packages loaded, 0 targets configured).
INFO: Found 1 target...
Target //:bin up-to-date:
  bazel-bin/bin
INFO: Elapsed time: 3.119s, Critical Path: 3.07s
INFO: 3 processes: 1 internal, 2 linux-sandbox.
INFO: Build completed successfully, 3 total actions
chiwang@chiwang ~/d/bazel-day-demo>
```

END

THANK YOU

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